

A Markov Decision Process (MDP) Formulation for Traffic Signal Control with Multi-Source State Sampling

1. Problem Overview

We consider a traffic signal control problem where a reinforcement learning (RL) agent aims to reduce risky behaviors such as red-light running.

At each decision step, the agent observes traffic conditions through **multiple information sources**, encodes them, and selects signal control actions.

The problem is modeled as a **Markov Decision Process (MDP)** to clearly define the environment dynamics, observations, and reward mechanism.

2. MDP Definition

The traffic control problem is formulated as an MDP:

$$M = (S, A, P, R, \gamma, \rho_0)$$

where:

- S is the set of environmental states,
 - A is the action space,
 - P is the state transition function,
 - R is the reward function,
 - γ is the discount factor,
 - ρ_0 is the initial state distribution.
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3. Environmental State

The true environment state at time step t is denoted as $s_t \in S$, which includes both macroscopic and microscopic traffic information:

$$s_t = \{x_t, p_t, \tau_t, \mathcal{V}_t\}$$

where:

- x_t : macroscopic traffic variables (lane density, queue length, average speed),
- p_t : current traffic signal phase and remaining time,
- τ_t : temporal context (time of day, cycle index),
- $\mathcal{V}_t$: vehicle-level information (positions, speeds, driver behavior).

This full state is internal to the simulator and may not be directly observable by the agent.

4. Multi-Source Observation and Sampling

At each time step, the agent receives a **sampled observation** composed of three components:

$$O_t = \{b_{1,t}, b_{2,t}, b_{3,t}\}$$

Each component corresponds to a different information source:

b_1 — Traffic State

A structured snapshot of the current traffic condition:

$$b_{1,t} = f_{\text{traffic}}(s_t)$$

It includes lane densities, queue lengths, average speeds, and signal phase indicators.

b_2 — Temporal Context

Time-related information that affects traffic patterns:

$$b_{2,t} = f_{\text{time}}(s_t)$$

Examples include time-of-day encoding, cycle position, or peak/off-peak indicators.

b_3 — Sequence History

Historical information capturing temporal dependencies:

$$b_{3,t} = f_{\text{hist}}(s_{t-H:t}, a_{t-H:t-1})$$

This typically consists of recent traffic states and signal actions over a fixed time window.

Together, $\{b_1, b_2, b_3\}$ form the agent's observation at each step.

5. Action Space

The agent selects a traffic signal control action $a_t \in \mathcal{A}$

Actions can be defined as:

- **Discrete:** selecting the next signal phase, or
 - **Continuous:** adjusting green-light durations.
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6. State Transition

The environment evolves according to:

$$s_{t+1} \sim P(\cdot \mid s_t, a_t)$$

State transitions are governed by a traffic simulator that models vehicle arrivals, driving behavior, and signal control effects, including stochastic disturbances.

7. Reward Function (Task 1)

The target task is to **maximize the average time interval between red-light-running accidents**.

Let $E_t \in \{0, 1\}$ indicate whether a red-light-running accident occurs at time t .

Let Δ_t denote the time elapsed since the last accident.

A practical reward definition is:

$$r_t = \Delta_{t+1} - \Delta_t - \lambda E_t$$

This formulation:

- rewards longer accident-free intervals,
 - penalizes the occurrence of accidents,
 - provides a dense learning signal suitable for RL.
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8. Episodes and Initialization

Each episode represents a fixed traffic control horizon (e.g., 30–60 minutes).

The initial state $s_0 \sim \rho_0$ is sampled from diverse traffic conditions, including different demand levels and temporal contexts.

9. Relevance to Encoder-Based RL

In this MDP:

- The **ground-truth reward** depends on all three sources $\{b_1, b_2, b_3\}$,

- Different encoders may be used to process each source,
- The RL model learns from encoded observations to estimate rewards and optimize control policies.

This formulation provides a clear and interpretable foundation for studying **multi-source state encoding and representation learning in traffic signal control**.